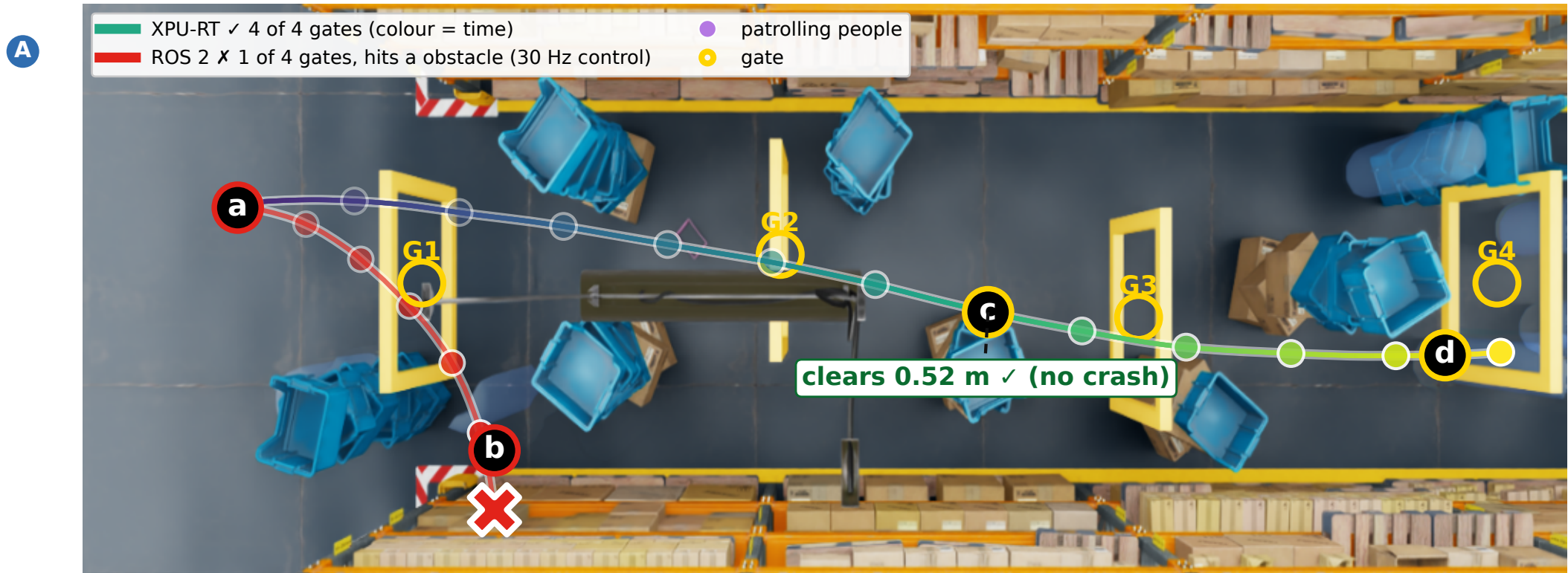
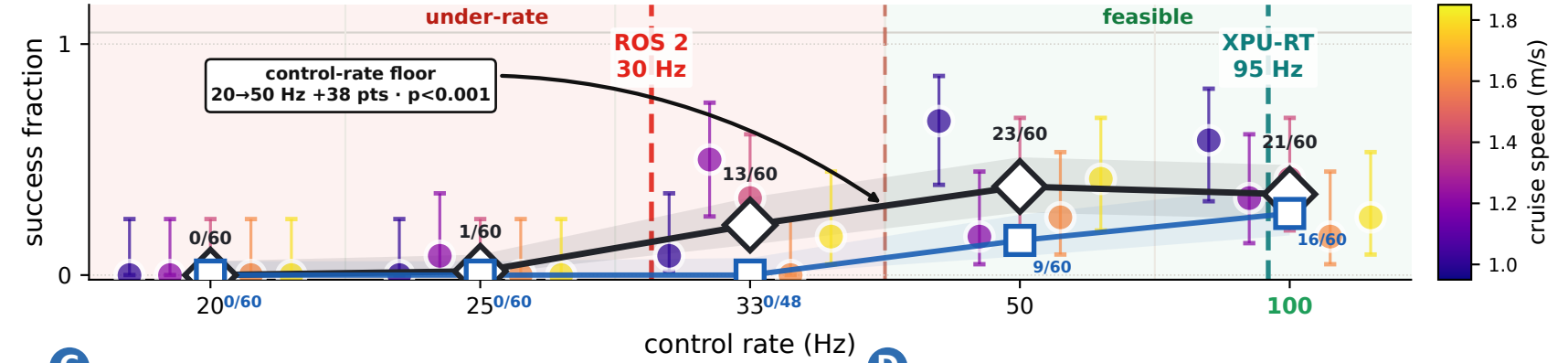


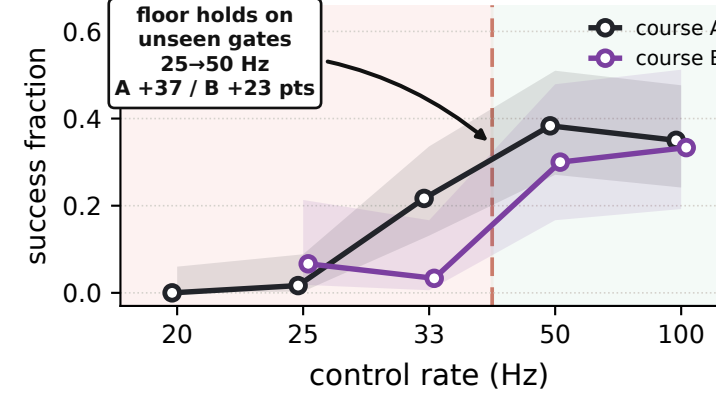
Warehouse gate-course showdown — same scene and drone, gain, at 1.8 m/s
XPU-RT · CP-SAT (95 Hz control) clears all 4 gates
ROS 2 · two model instances on all 8 cores (30 Hz control) clears G1 → hits a obstacle
one seed of this scene (seed 1001); over its 12 seeds XPU-RT completes 2 and ROS 2 none



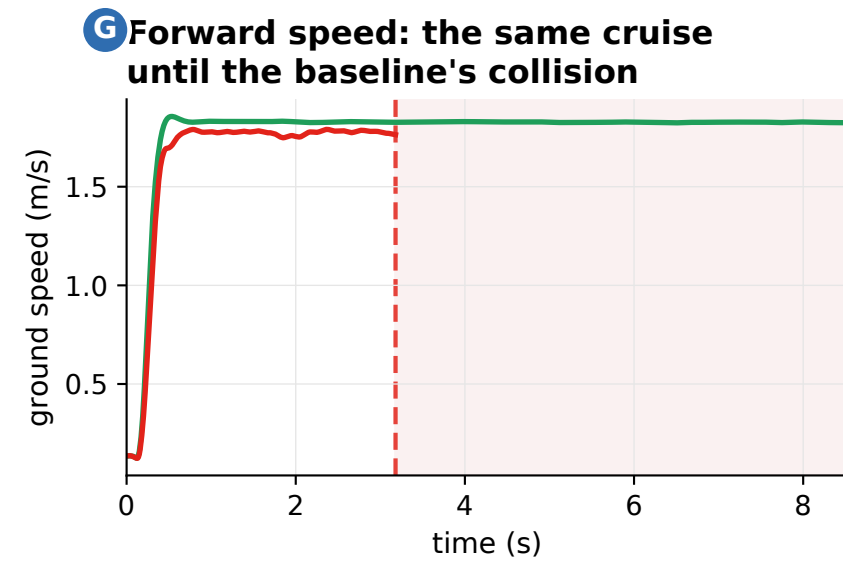
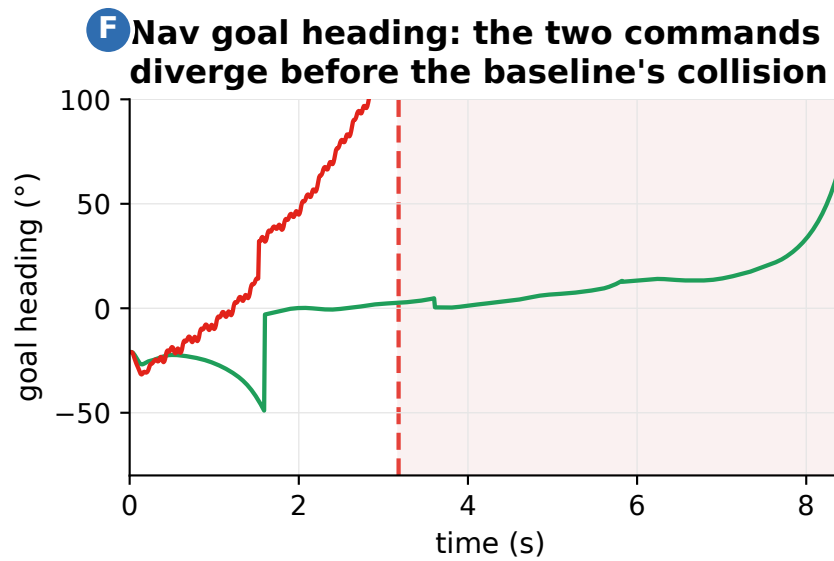
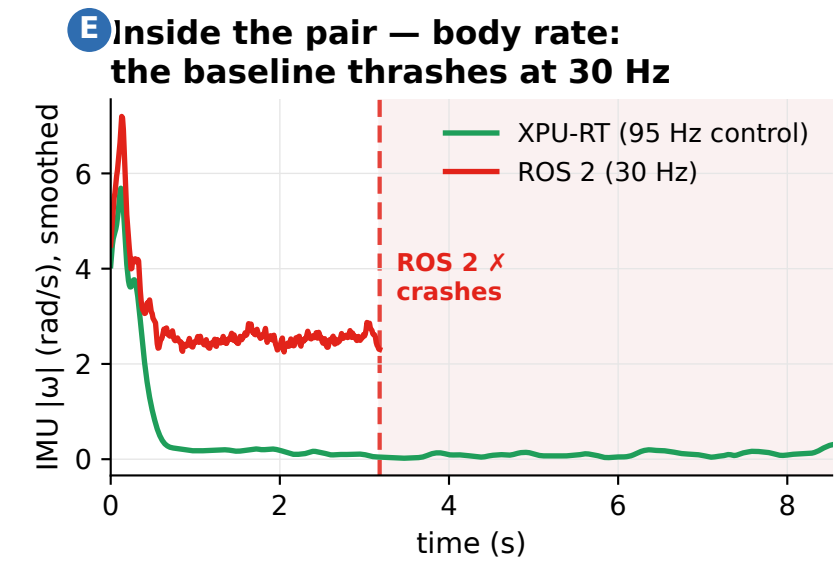
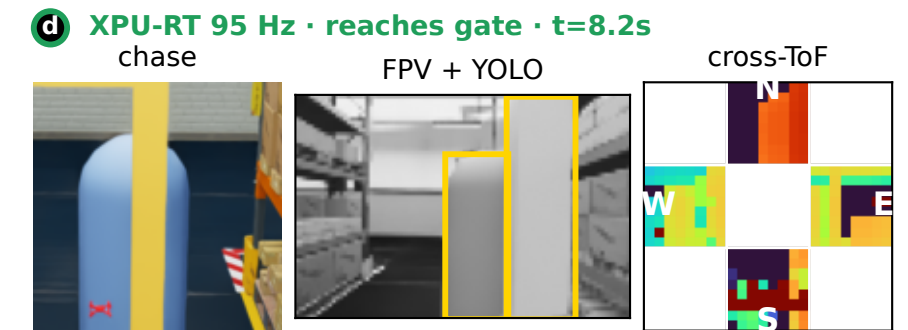
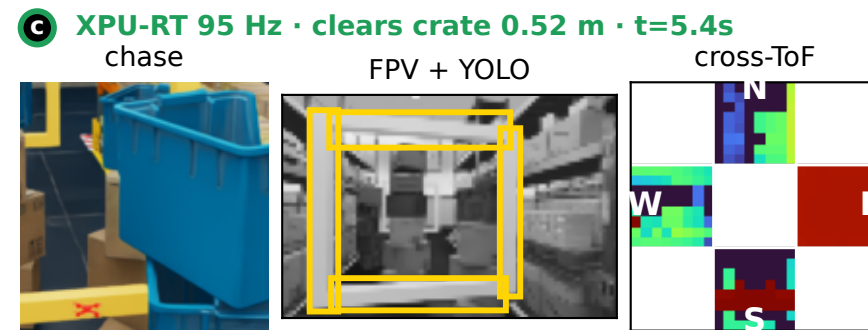
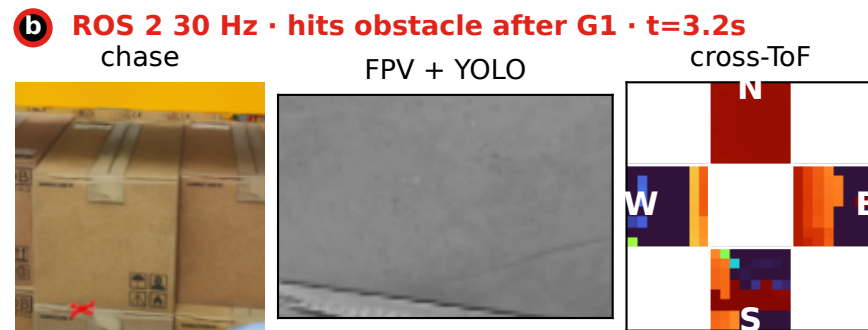
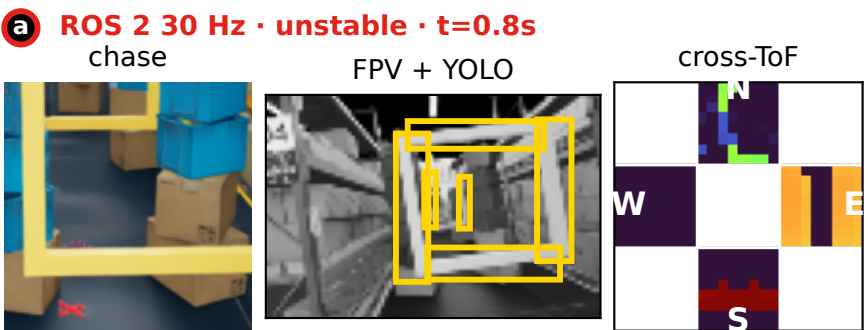
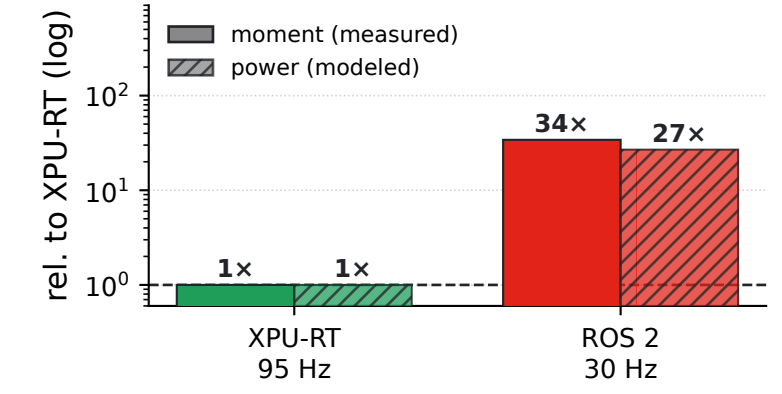
B Control-rate envelope (rate injected, 300 flights): success against control rate — ROS 2's 30 Hz and XPU-RT's 95 Hz marked



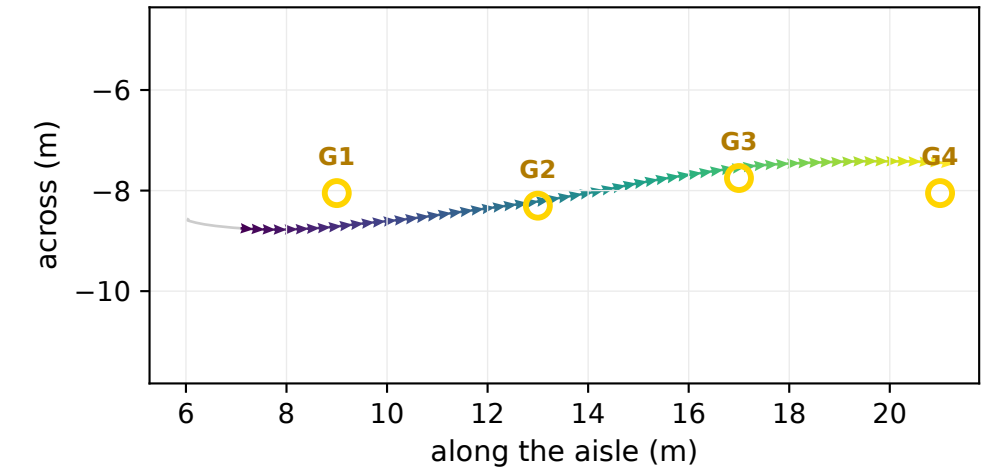
C Generalization: unseen gates (course B)



D Mechanism: the baseline thrashes (n=6 flights/arm)



H XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



I Onboard K1 schedule, measured at the 30 Hz camera — XPU-RT · CP-SAT places the kernels over all 8 harts: camera→control 55 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, control on 7 harts, nav on 7 harts
ROS 2 · all 8 cores (5 processes · single executor · chained control · unpinned): camera→control 31 ms, control every 33.3 ms, 0/298 frames late · YOLO on 4 harts, nav on 1 hart, control on 1 hart

